

E-Commerce Product Analytics

User behavior, conversion, retention & experimentation

Executive Summary

This project analyzes user interaction data from an e-commerce platform to understand how visitors progress through the purchasing funnel, how they engage with the platform, and how user behavior changes over time.

The analysis covers funnel conversion, session behavior, daily and weekly engagement, conversion trends, cohort retention, and statistical experimentation methodology.

The dataset contains **115,403 unique visitors**, **214,787 events**, and **65,173 unique products** across approximately 10 days of activity.

Metric	Result
Unique Visitors	115,403
Total Events	214,787
Unique Products	65,173
Inferred Sessions	133,336
View → Cart	2.55%
View → Purchase	0.80%

Key Findings

- Only **2.55%** of product viewers proceeded to add an item to their cart.
- Among users who added products to their cart, **31.36%** completed a purchase.
- Average daily active users were approximately **11,352**, with a peak of **13,789**.
- Week-1 to Week-2 retention for the first cohort was **3.39%**.
- The simulated A/A test produced a **p-value of 0.687**, providing insufficient evidence of a difference between the randomly assigned groups.

User Funnel Analysis

Funnel Stage	Unique Users
Viewed Product	1,404,179
Added to Cart	37,722
Purchased	11,719

Product-view to cart conversion was 2.55%, while cart-to-purchase conversion was 31.36%. Overall view-to-purchase conversion was 0.80%.

Session Analysis

User activity was grouped into sessions using a **30-minute inactivity threshold**. This transformed the event-level data into **133,336 inferred user sessions**.

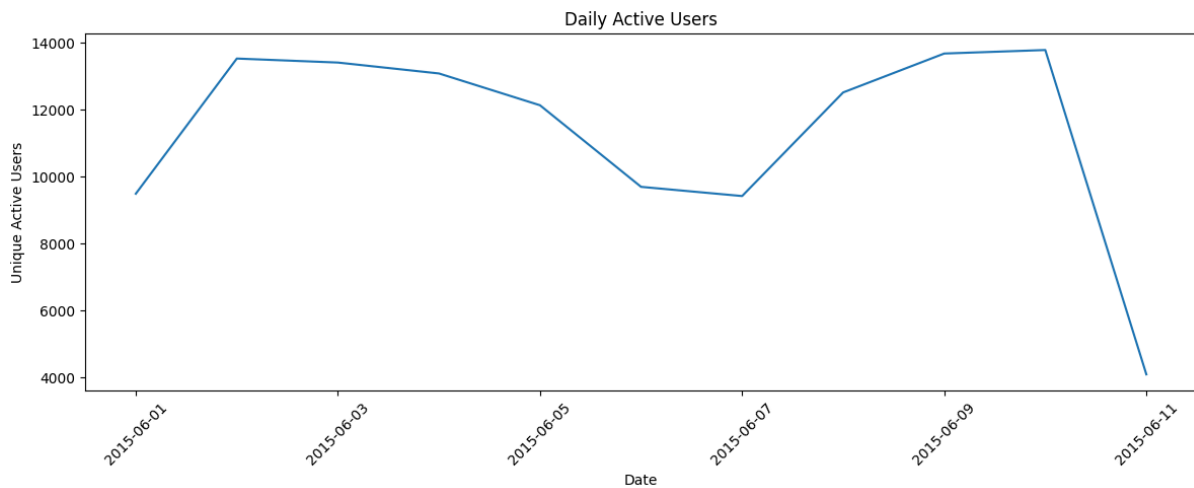
Sessionization provides a more useful unit for behavioral analysis than individual events, allowing user activity to be examined as continuous visits rather than isolated interactions.

The session-level analysis was used to distinguish browsing behavior from purchasing behavior and support subsequent conversion analysis. Because session boundaries are inferred from inactivity rather than recorded directly by the platform, the results depend on the chosen 30-minute threshold.

Session Metric	Result
Inferred Sessions	133,336
Session Definition	30-minute inactivity threshold

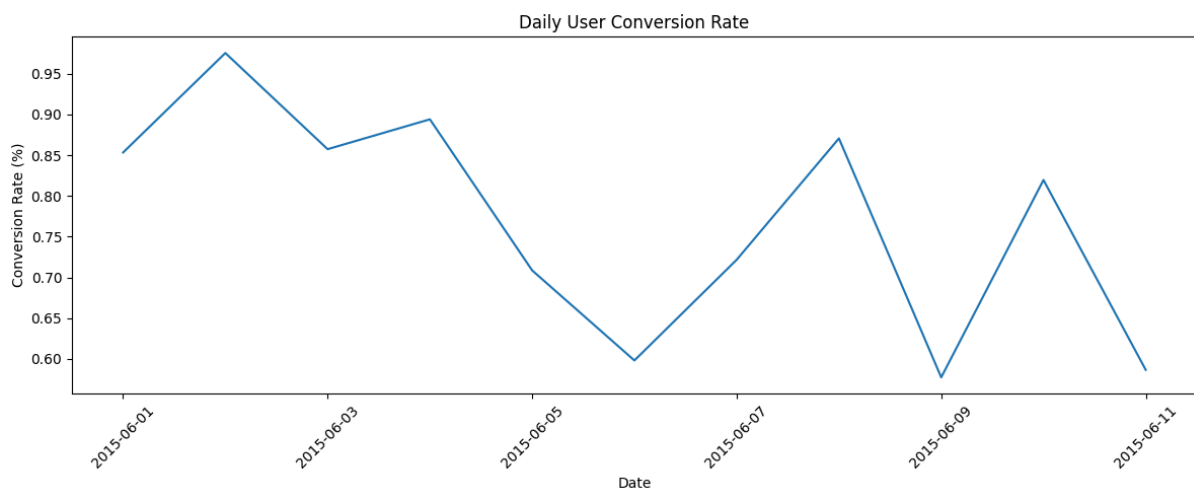
Engagement & Conversion

Daily Active Users



Daily active users averaged approximately 11,352 during the observed period, with peak daily activity reaching 13,789 users.

Daily Conversion Rate



Conversion was analyzed at the daily level to examine how purchase behavior varied throughout the observation period. Because the dataset covers only approximately 10 days, the analysis should be interpreted as a short-term behavioral snapshot rather than a long-term trend.

Cohort & Experimentation

Cohort Retention

Metric	Result
First cohort Week-1 → Week-2 retention	3.39%
Dataset observation period	~10 days

Users were grouped according to their first recorded activity week. The first cohort showed 3.39% retention into the following week. Because the dataset spans only approximately 10 days, there is insufficient follow-up data for a comprehensive long-term cohort analysis.

A/A Experimentation

Metric	Result
Control conversion	0.8095%
Treatment conversion	0.7884%
Absolute difference	-0.0211 pp
Relative lift	-2.61%
p-value	0.6871
95% CI	-0.1238 to +0.0816 pp

Because the source dataset does not contain actual experimental treatment assignments, an **A/A simulation** was used to demonstrate the experimentation workflow. Users were randomly assigned to two groups and purchase conversion was compared.

The observed difference was not statistically significant (**p = 0.687**). The confidence interval also included zero, so the analysis does not provide sufficient evidence of a difference between the two randomly assigned groups.

Findings & Limitations

Key Findings

1. Funnel drop-off

The largest observed funnel drop occurs between product viewing and cart addition, with only 2.55% of viewers reaching the cart stage.

2. Stronger downstream conversion

Once users added an item to their cart, 31.36% proceeded to purchase, indicating substantially higher conversion at the later funnel stage.

3. Short-term engagement

The platform generated 11,352 average daily active users, with daily engagement varying during the observation window.

4. Experimentation

The simulated A/A test produced no statistically significant difference, demonstrating how conversion experiments can be evaluated using effect size, p-values, and confidence intervals.

Limitations

- The dataset covers approximately 10 days, limiting long-term trend and retention analysis.
- The dataset contains observational user behavior rather than genuine randomized experiments.
- The A/A test is simulated and should not be interpreted as evidence about an actual product change.
- Session boundaries are inferred using a 30-minute inactivity threshold.
- Observed correlations and behavioral patterns do not establish causation.

Conclusion

This project demonstrates an end-to-end product analytics workflow, from raw event-level data through user funnel analysis, sessionization, engagement measurement, cohort retention, and statistical experimentation. The analysis highlights both measurable product behavior and the limitations that analysts must consider when interpreting observational data.